

DEPI Graduation Project

The Analytix Team

Task1: Build Data Model, Data Cleaning and Preprocessing:

Dataset Link :

[[https://drive.google.com/file/d/1c67mQ\\$XNTcpED9MbkbB_qIFMErMoAExFk/view?usp=drive_link](https://drive.google.com/file/d/1c67mQ$XNTcpED9MbkbB_qIFMErMoAExFk/view?usp=drive_link)]

Notion Link : [<https://www.notion.so/Superstore-Sales-DEPI-Graduation-Project-19e07a7c03c1808f82a9e345a8dcd816?pv=4>]

Data Cleaning and Preprocessing

1. Importing Libraries

- Loads essential Python libraries like pandas, numpy, and plotly.express for data manipulation and visualization.

2. Loading the Dataset

- Reads the Superstore dataset from an Excel file into a Pandas DataFrame.

3. Initial Data Exploration

- Displays the first few rows.
- Checks dataset structure (info()).
- Identifies duplicate rows and missing values.
- Displays rows containing missing values.

4. Outlier Detection

- Creates box plots for 'Sales' and 'Profit' using plotly.express to visualize outliers.

5. Handling Missing Values

- 'Sales' & 'Profit' → Filled with median to avoid outlier influence.
- 'Quantity' & 'Discount' → Filled with mean.

6. Data Type Conversion

- Converts 'Quantity' to integer (as it represents whole numbers).
- Converts 'Order Date' & 'Ship Date' to datetime format for proper date handling.



8. Final Data Verification

- Displays dataset information after modifications.
- Ensures there are no missing values left.

Deliverables:

o Data preprocessing notebook :

[https://drive.google.com/file/d/1x3N1OLs_xdqB5vqXfSczmll4c-ophaWb/view?usp=drive_link]

o Cleaned dataset ready for analysis :

[https://docs.google.com/spreadsheets/d/1ykW8aAtnkwJtDyZShH67nT7nsCvQjbWf/edit?usp=drive_link&ouid=112242095008648923771&rtpof=true&sd=true]

o SQL query : [https://drive.google.com/file/d/1n8ew8ft-uf9lSl9-5PG19GBBEsLRhMt5/view?usp=drive_link]

Build A Data Model



The **Star Schema** is a data modeling approach designed for efficient querying and reporting. It consists of:

- ✓ **Fact Table** – Stores transactional data (e.g., sales, quantity, profit)
- ✓ **Dimension Tables** – Provide descriptive attributes (e.g., customers, products, dates)

Converting Superstore Data to Star Schema

1. Fact Table: Fact_Sales (Central Table)

- o Contains sales transactions, including sales amount, quantity, discount, and profit.
- o Has foreign keys linking to dimensions: customers, products, ship mode, and dates.

2. Dimension Tables (Lookup Tables for Filtering & Grouping):

- o **Dim_Customers:** Stores customer details (ID, name, segment, location).
- o **Dim_Products:** Contains product info (ID, name, category, sub-category).
- o **Dim_ShipModes:** Stores shipping mode types.



- **Dim_Dates:** Provides date-based attributes (year, month, day) for time analysis.

🚀 Why Use a Star Schema?

- ✓ **Faster Queries:** Reduces complex joins, improving performance.
- ✓ **Better Readability:** Organized structure for easy analysis.
- ✓ **Scalability:** Suitable for large datasets & BI tools like Power BI/Tableau.

This schema enables efficient sales analysis, customer segmentation, and trend forecasting 🚀💡

